Deep Learning – Part-01:

Thursday, September 10, 2026 6:56 PM

Deep Learning — From Fundamentals to Interview Level

Deep Learning is a subfield of Machine Learning that uses **neural networks with multiple layers** to learn patterns directly from data.

A useful mental model is:

> **Input → Weighted Sum + Bias → Activation → Prediction → Loss → Backpropagation → Weight Update → Repeat**

Once you understand that loop, most neural-network concepts become much easier.

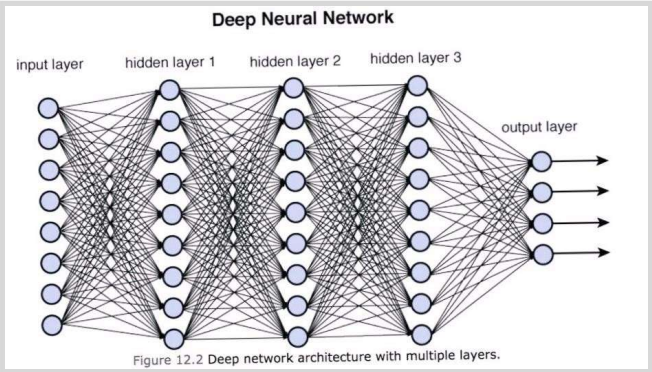
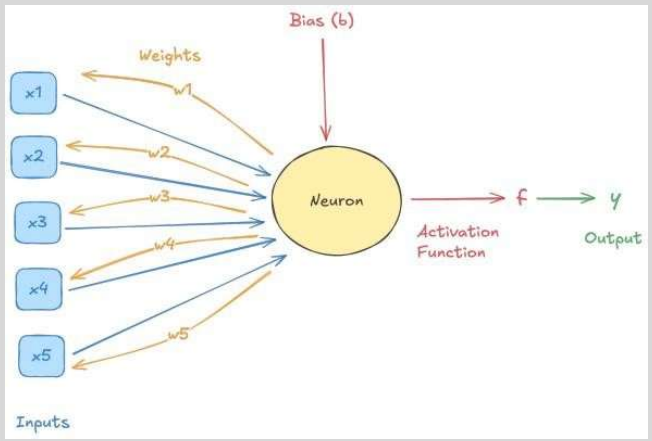
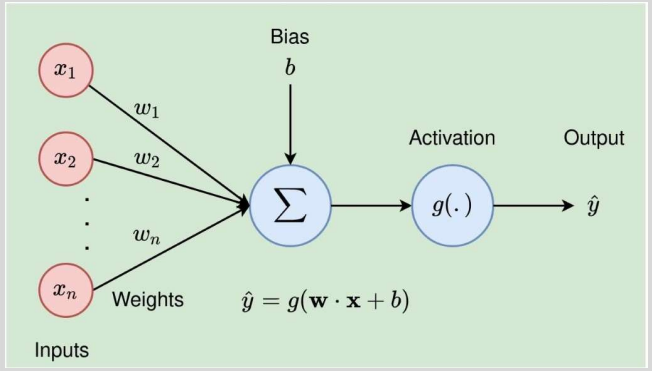


Figure 12.2 Deep network architecture with multiple layers.





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1. What is Deep Learning?

A traditional ML model often relies heavily on **feature engineering**.

For example, suppose we want to predict whether an image contains a cat.

A traditional approach might require manually extracting:

- edges
- colors

- shapes
- textures
- object dimensions

Deep Learning can learn these representations automatically.

For an image, a deep network might learn approximately:

Pixels
↓
Edges
↓
Simple shapes
↓
Parts of objects
↓
Complete objects
↓
"Cat"

This is why deep learning is particularly powerful for:

- Computer Vision
- Natural Language Processing
- Speech recognition
- Recommendation systems
- Time-series problems
- Generative AI
- Large Language Models

A **Deep Neural Network (DNN)** consists of multiple layers of neurons. The layers between input and output are called **hidden layers**.

Why "deep"?

Consider:

Input
↓
Hidden Layer 1
↓
Hidden Layer 2
↓
Hidden Layer 3
↓
Hidden Layer 4
↓
Output

More hidden layers → greater **depth**.

The number of neurons within a layer is its **width**.

2. The Most Important Equation

Almost everything begins with this equation:

$$z = w_1x_1 + w_2x_2 + \cdots + w_nx_n + b$$

or in vector notation:

$$z = W^TX + b$$

Then:

$$a = f(z)$$

where:

Symbol	Meaning
X	Input
W	Weights
b	Bias
z	Weighted sum / pre-activation
f	Activation function
a	Activation/output

This basic neuron computation is the foundation of feed-forward neural networks.

3. Inputs

Definition

An **input** is a feature/value supplied to the neural network.

For example, suppose we want to predict house prices:

x_1 = Area
x_2 = Number of bedrooms
x_3 = Age of house
x_4 = Location score

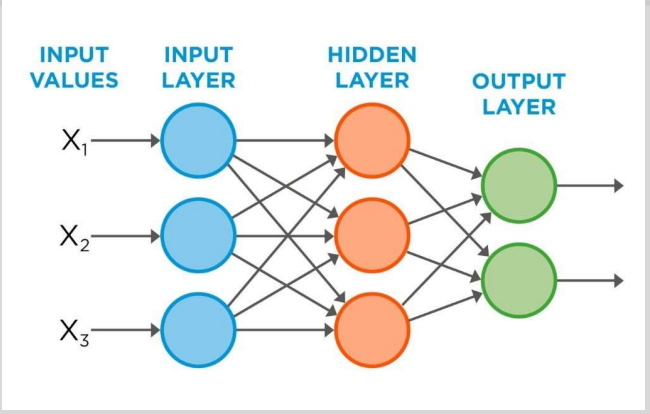
These are the inputs.

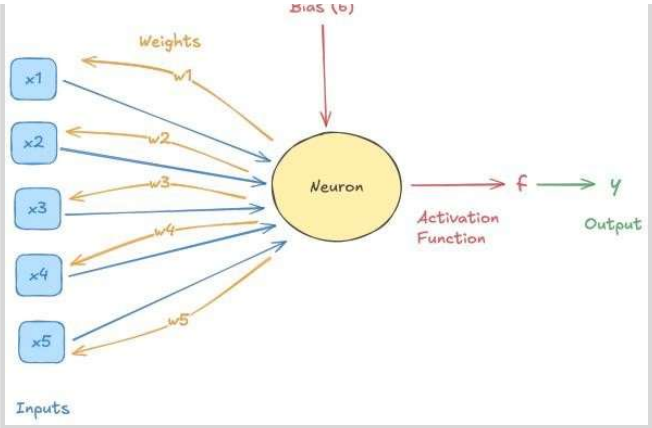
For your **coronary heart disease project**, inputs could be:

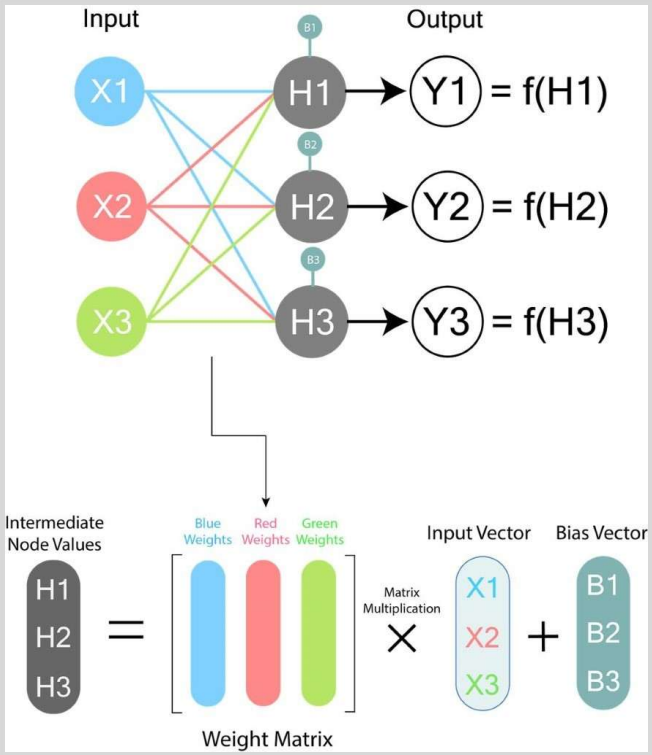
Age
Gender
Education
Smoking
Cigarettes/day
Blood pressure
BMI
Glucose
Cholesterol
...

The input vector could therefore be:

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}$$







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Important interview point

An **input feature is not necessarily a neuron in the biological sense**.

In an artificial neural network, the input layer essentially represents the supplied feature values.

4. Weights

Definition

A **weight** determines how strongly an input influences a neuron.

Suppose:

$$z = 5x_1 + 0.2x_2$$

Here:

Weight of $x_1 = 5$
Weight of $x_2 = 0.2$

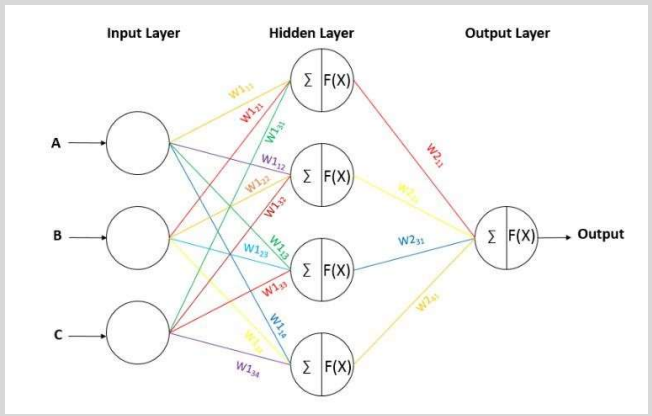
The neuron considers much more strongly.

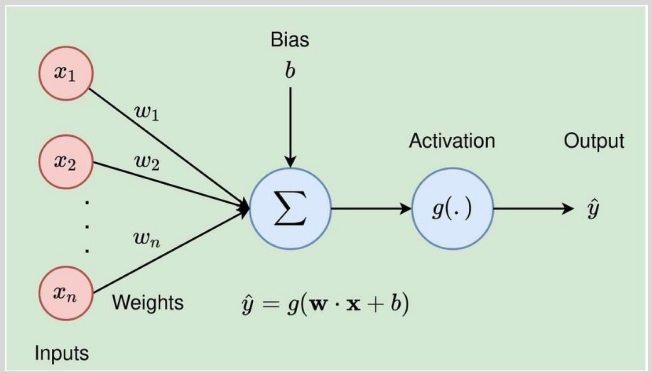
$$\text{The neuron considers } x_1 \text{ much more strongly.}$$

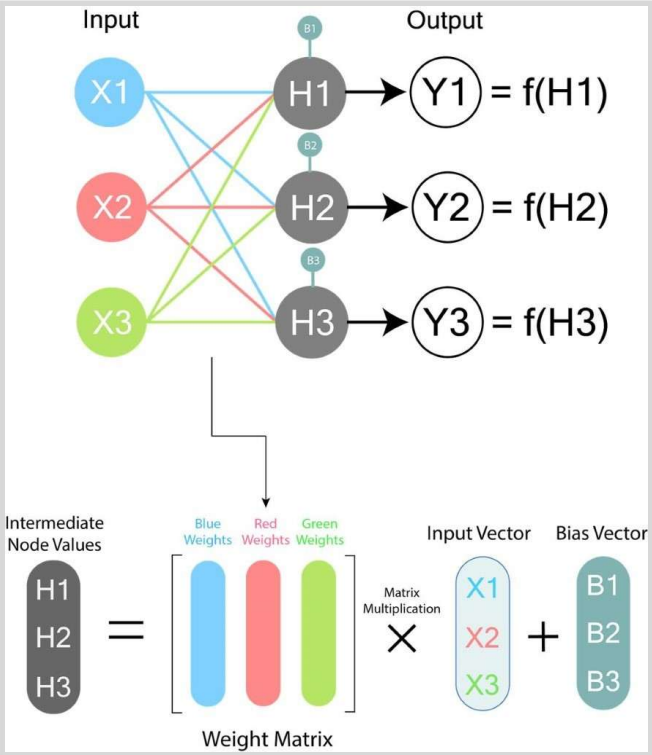
A positive weight means the input pushes the activation upward.

A negative weight means the input pushes it downward.

A weight close to zero means the input has relatively little influence.







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Interview question

Q: Are weights learned or manually assigned?

A: Learned during training using optimization algorithms such as gradient descent and its variants.

5. Bias

Definition

A **bias** is an additional learnable parameter added to the weighted sum.

$$z = \sum_i w_i x_i + b$$

Think of it as a **shift/offset**.

Without bias:

$$z = w_1 x_1 + w_2 x_2$$

With bias:

$$z = w_1 x_1 + w_2 x_2 + b$$

The bias allows the neuron to shift its activation threshold rather than forcing the decision boundary through the origin.

Simple intuition

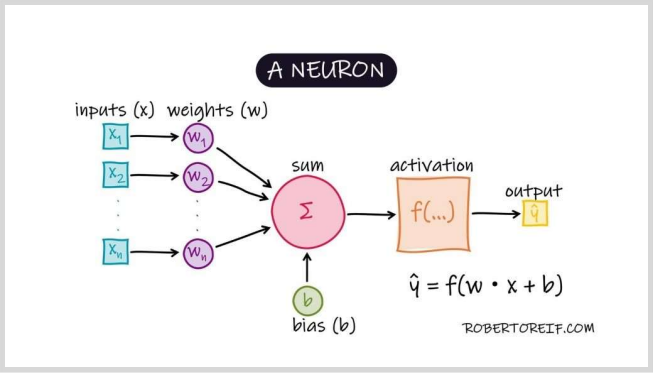
Suppose:

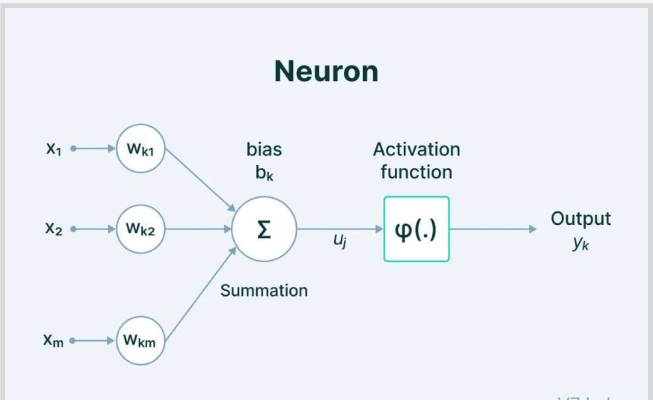
$$z = 2x - 5$$

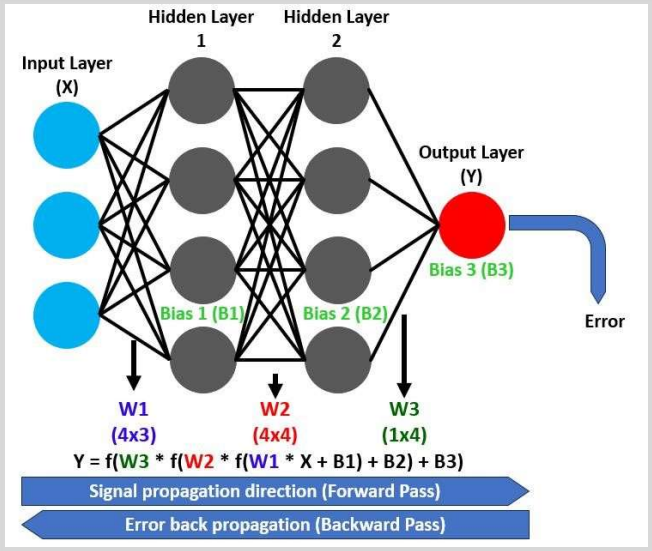
The -5 changes when the neuron starts activating.

So:

> **Weights control influence; bias controls the offset.**







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Interview question

Q: Why do we need bias?

Without bias, the neuron has reduced flexibility because its weighted linear transformation is constrained to pass through the origin.